

# Abu Henaf Rashid Ahmad Mufti

Computer Science Undergraduate | AI/ML Engineering & LLM Applications
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**LinkedIn** | **GitHub** | **Portfolio**

## Professional Profile

Final-year Computer Science and Engineering student at United International University, majoring in Data Science, with hands-on experience in AI/ML engineering, LLM applications, AI agents, RAG systems, and software development. Experienced in Python, LangGraph, FastAPI, machine learning, and AI API integration. Uses Claude Code and Antigravity primarily for AI-assisted development, including implementation, debugging, and integration. Co-first author of a peer-reviewed ACM publication, with additional research manuscripts under review.

## Education

**United International University** *Dhaka, Bangladesh*
*Bachelor of Science in Computer Science and Engineering* | Final Year | Major: Data Science
**CGPA:** 3.51 / 4.00

**Academia School: IAL (A Level), Edexcel Curriculum** 2019 – 2021
**Academia School: IGCSE (O Level), Edexcel Curriculum** 2018 – 2019
*Edexcel Award, Daily Star Award*

## Experience

**Student Researcher** | Elite Research Labs **September 12, 2026 – Present**
- Structure research pipelines and implement the code for AI/ML experiments.
- First author on a journal manuscript in preparation for Transaction in Image Processed; responsible for pipeline design and coding.

## Technical Skills

**Programming:** Python, JavaScript, TypeScript, Java, SQL
**AI/ML:** Machine Learning, Computer Vision, Explainable AI, RAG, LLM Applications, AI Agents
**Frameworks & Libraries:** LangGraph, FastAPI, Scikit-learn, Pandas, NumPy, OpenCV
**Web Development:** React.js, Next.js, Express.js, HTML, CSS, Tailwind CSS
**Databases & APIs:** SQL, Database Design, REST API Integration
**AI Coding Tools:** Claude Code, Antigravity, Cursor, Trae, Qoder, Kilo
**Tools:** Git, GitHub, Roboflow, Streamlit, Vercel

## Selected AI & Software Projects

**Data Seeker** *Natural Language to SQL* **GitHub** | **Live**
- Developed a natural language interface for converting user queries into validated SQL queries and analyst-style summaries.
- Implemented query validation and automated analytical reporting. Built with LLMs, SQL, OpenRouter, Plotly, and Streamlit.

**Web ScrapChat** *AI Agent Application* **GitHub**
- Built a full-stack search-chat application with a Next.js frontend and streaming, LangGraph-orchestrated tool calls over SSE.
- Stack: Next.js, FastAPI, LangGraph, SSE, Tailwind CSS.

**Research Assist** *AI Research Agent* **GitHub** | **Live**
- Built a stateful LangGraph agent that searches arXiv, reads papers, and produces publication-style summaries.
- Stack: LangGraph, Gemini, arXiv, Streamlit.

**Unified MediChat** *AI Healthcare Assistant* **GitHub**
- Developed an AI-powered healthcare assistant with RAG, voice, and vision workflows, integrating web and academic search and multiple information sources.
- Stack: FastAPI, RAG, LangGraph, Groq, Streamlit.

**NoteDex** *AI-Powered Note-Taking Platform* **GitHub** | **Live** | **Demo**
- Built a full-stack AI note-taking platform using Next.js, React, TypeScript, Tailwind CSS, Express.js, MongoDB, and Google Gemini.
- Implemented AI-powered features with reusable component architecture and mobile-first responsive layouts; deployed on Vercel.

## PUBLICATION

**An Explainable and Scalable Framework for Compost Maturity Prediction**
Co-first author | Peer-reviewed conference paper | ACM Digital Library, 2024 | **Status: Published**
**View on ACM Digital Library**

## ONGOING / UNDER-REVIEW RESEARCH

**DNA Protein-Site Binding Evaluation** *Status: Manuscript finalized and under review*
- Research involving biological sensorgram data and feature engineering for protein-site binding evaluation.
- First author. Main role: pipeline, prompt engineering, context, and model evaluation.

**Pneumonia Lung Disease Classification** *Status: Manuscript finalized and under review*
- Research involving chest X-ray preprocessing, segmentation, and deep-learning classification.
- First author. Main role: pipeline, prompt engineering, context, and model evaluation.

**A Multimodal Risk Assessment and Monitoring Framework for Assisted Elderly Living** *Status: Journal manuscript in preparation*
- Elderly-behavior video annotation, dataset curation, and annotation quality control using Roboflow.
- Main role: pipeline, prompt engineering, context, and model evaluation.

**AQI Prediction and Generalisation** *Status: Ongoing research*
- Air-quality prediction research exploring Random Forest, XGBoost, CatBoost, and SHAP.
- First author. Main role: pipeline, prompt engineering, context, and model evaluation.

**Shadhu Bangla OCR and NLP Framework** *Status: Journal manuscript in preparation*
- Historical Bangla manuscript OCR dataset preparation, annotation, augmentation, and quality control.
- First author. Main role: pipeline, prompt engineering, context, and model evaluation.

## Certifications & Achievements

- Accomplished Competitor, University Physics Competition (UPC) 2025, Ideation and Methodology.
- Peer-reviewed ACM Digital Library publication, 2024 (see Publication above).
- **Anthropic Education — AI Fluency: Framework & Foundations**
- **Anthropic Education — AI Capabilities and Limitations**
- **freeCodeCamp — Machine Learning with Python**
- **freeCodeCamp — Data Analysis with Python**
- **HackerRank — SQL (Advanced)**

## Career Focus

**AI Engineering | Machine Learning | LLM Applications & AI Agents | Data Science | Backend Engineering | Applied Research**

Seeking a full-time AI/ML engineering role focused on building reliable AI-powered products and practical machine-learning systems.